

Week 4 Deliverable: Baseline Rover Geometry, Mass Budget, and First CAD Block Model

Omkar Shende

June 16, 2026

1 Objective

The objective of Week 4 is to convert the lunar regolith transport rover from a concept-level design into a preliminary engineering baseline. This deliverable finalizes the baseline payload, estimates dump bed volume, defines first-pass rover dimensions, estimates dry mass, creates a mass budget, defines initial rocker-bogie geometry, and begins center-of-gravity reasoning.

The rover is intended to transport lunar regolith from a collection zone to a habitat shielding zone. The design uses a six-wheel rocker-bogie mobility system, a rear dump bed, and a ruggedized chassis suitable for early lunar surface construction support.

2 Baseline Payload

The baseline payload is finalized as:

$$m_{payload} = 340 \text{ kg} \tag{1}$$

This corresponds approximately to:

$$340 \text{ kg} \approx 750 \text{ lb} \tag{2}$$

This payload represents the amount of lunar regolith the rover is designed to transport per trip.

3 Dump Bed Volume Estimate

The preliminary dump bed volume is selected as:

$$V_{bed} \approx 0.30 \text{ m}^3 \tag{3}$$

Using the payload mass and bed volume, the implied average bulk density is:

$$\rho = \frac{m}{V} \quad (4)$$

$$\rho = \frac{340 \text{ kg}}{0.30 \text{ m}^3} \quad (5)$$

$$\rho \approx 1133 \text{ kg/m}^3 \quad (6)$$

This is reasonable as an initial estimate for loose lunar regolith storage because the dump bed does not need to assume fully compacted material. The volume is also manageable for a first block model.

A possible rectangular dump bed size is:

$$L_{bed} = 1.2 \text{ m} \quad (7)$$

$$W_{bed} = 0.75 \text{ m} \quad (8)$$

$$H_{bed} = 0.35 \text{ m} \quad (9)$$

The resulting volume is:

$$V = L_{bed}W_{bed}H_{bed} \quad (10)$$

$$V = (1.2)(0.75)(0.35) \quad (11)$$

$$V = 0.315 \text{ m}^3 \quad (12)$$

Therefore, the first CAD block model will use a dump bed volume of approximately:

$$V_{bed} = 0.30 \text{ m}^3 \text{ to } 0.32 \text{ m}^3 \quad (13)$$

4 Preliminary Rover Dry Mass Estimate

The dry mass is the mass of the rover without the regolith payload. A preliminary subsystem mass estimate is shown in Table 1.

Table 1: Preliminary rover dry mass estimate

Subsystem	Estimated Mass
Chassis/frame	90 kg
Rocker-bogie suspension	70 kg
Six wheels and hubs	60 kg
Motors and gearboxes	45 kg
Dump bed and actuator	80 kg
Battery system	120 kg
Solar panels and charging hardware	35 kg
Sensors, antennas, and computer	25 kg
Electronics enclosure and dust protection	45 kg
Fasteners, brackets, wiring, and margin	80 kg
Estimated Dry Mass	650 kg

Therefore, the preliminary dry mass is:

$$m_{dry} \approx 650 \text{ kg} \quad (14)$$

The loaded rover mass is:

$$m_{loaded} = m_{dry} + m_{payload} \quad (15)$$

$$m_{loaded} = 650 + 340 \quad (16)$$

$$m_{loaded} = 990 \text{ kg} \quad (17)$$

For early design calculations, the loaded mass can be rounded to:

$$m_{loaded} \approx 1000 \text{ kg} \quad (18)$$

5 Mass Budget

The overall mass budget is split into dry rover mass and useful payload mass.

Table 2: Preliminary loaded rover mass budget

Category	Mass
Dry rover mass	650 kg
Regolith payload	340 kg
Total loaded mass	990 kg

The payload fraction is:

$$f_{payload} = \frac{m_{payload}}{m_{loaded}} \quad (19)$$

$$f_{payload} = \frac{340}{990} \quad (20)$$

$$f_{payload} \approx 0.34 \quad (21)$$

Therefore, approximately:

$$f_{payload} \approx 34\% \quad (22)$$

of the loaded rover mass is useful payload.

This is a reasonable starting point for a rugged lunar construction-support rover because the rover also requires a strong chassis, suspension, drivetrain, battery, electronics, sensors, and dust-protected systems.

6 Wheelbase and Track Width Estimate

The rover geometry is sized around the payload bed, rocker-bogie suspension, and stability requirements.

A preliminary range of rover dimensions is shown in Table 3.

Table 3: Preliminary rover dimension ranges

Dimension	Estimated Range
Overall length	2.6 m to 3.0 m
Wheelbase	2.0 m to 2.3 m
Track width	1.6 m to 1.9 m
Wheel diameter	0.45 m to 0.60 m
Dump bed length	1.2 m
Dump bed width	0.75 m
Dump bed height	0.35 m

The selected baseline dimensions for the first CAD block model are:

$$L_{rover} \approx 2.8 \text{ m} \quad (23)$$

$$W_{track} \approx 1.8 \text{ m} \quad (24)$$

$$D_{wheel} \approx 0.50 \text{ m} \quad (25)$$

The wheel radius is therefore:

$$r_{wheel} = \frac{D_{wheel}}{2} \quad (26)$$

$$r_{wheel} = \frac{0.50}{2} \quad (27)$$

$$r_{wheel} = 0.25 \text{ m} \quad (28)$$

These dimensions give the rover enough space for the dump bed while maintaining a stable width and realistic wheel spacing.

7 Rocker-Bogie Geometry

The rover uses a six-wheel rocker-bogie suspension system with three wheels on each side. Each side contains:

- One front wheel
- One middle wheel
- One rear wheel
- One main rocker arm
- One bogie link
- One central rocker pivot
- One differential connection between the left and right rocker arms

The purpose of the rocker-bogie system is to allow the rover to passively adapt to uneven terrain while keeping the rover body relatively stable.

The baseline rocker-bogie geometry for the first block model is shown in Table 4.

Table 4: Baseline rocker-bogie geometry

Geometry Feature	Baseline Estimate
Front-to-middle wheel spacing	0.9 m to 1.0 m
Middle-to-rear wheel spacing	0.9 m to 1.0 m
Total wheelbase	2.0 m to 2.3 m
Rocker pivot height	0.45 m to 0.65 m
Wheel radius	0.25 m
Ground clearance	0.30 m to 0.40 m

For the first CAD model, the following geometry should be used:

- Front wheel near the front of the chassis

- Middle wheel below or slightly behind the rocker pivot
- Rear wheel near the back of the chassis
- Dump bed centered between the left and right wheel sets
- Battery placed low in the chassis
- Electronics enclosure placed above or beside the battery
- Sensor mast or sensor package placed toward the front

The rocker-bogie system should first be modeled as simple bars, pivots, and wheel cylinders. Detailed bearings, joints, brackets, and motor housings can be added later.

8 Basic Lunar Weight Estimate

Although the rover has a large mass, its weight on the Moon is lower than it would be on Earth. Lunar gravitational acceleration is assumed to be:

$$g_{moon} = 1.62 \text{ m/s}^2 \quad (29)$$

Using the rounded loaded rover mass:

$$m_{loaded} = 1000 \text{ kg} \quad (30)$$

The loaded rover weight on the Moon is:

$$W_{moon} = m_{loaded}g_{moon} \quad (31)$$

$$W_{moon} = (1000)(1.62) \quad (32)$$

$$W_{moon} = 1620 \text{ N} \quad (33)$$

Assuming the load is distributed evenly across six wheels, the approximate normal load per wheel is:

$$N_{wheel} = \frac{W_{moon}}{6} \quad (34)$$

$$N_{wheel} = \frac{1620}{6} \quad (35)$$

$$N_{wheel} = 270 \text{ N} \quad (36)$$

This is a first-order estimate. In reality, the load on each wheel will change depending on slope, terrain, suspension articulation, and payload distribution.

9 Center-of-Gravity Thinking

Center of gravity is important because the rover must remain stable while:

1. Driving loaded
2. Driving unloaded
3. Climbing slopes
4. Turning
5. Dumping regolith
6. Crossing uneven terrain

The heaviest components are expected to be:

- Regolith payload
- Battery system
- Chassis/frame
- Dump bed
- Rocker-bogie suspension

The first design goal is to keep the center of gravity:

- Low
- Near the middle of the wheelbase
- Centered between the left and right wheels
- Inside the support polygon formed by the six wheel contact points

A preliminary longitudinal center-of-gravity assumption is:

$$x_{CG} \approx 0.50L_{wheelbase} \quad (37)$$

If the baseline wheelbase is:

$$L_{wheelbase} = 2.1 \text{ m} \quad (38)$$

then:

$$x_{CG} \approx 0.50(2.1) \quad (39)$$

$$x_{CG} \approx 1.05 \text{ m} \quad (40)$$

measured from the front wheel contact location.
A preliminary vertical center-of-gravity estimate is:

$$z_{CG} \approx 0.6 \text{ m to } 0.8 \text{ m} \quad (41)$$

above the ground.

The battery should be placed low in the chassis to reduce the vertical center of gravity. The dump bed should be centered over the frame rather than placed too far rearward. This helps reduce tipping risk while driving and dumping.

9.1 Preliminary Static Stability Check

A simple lateral tipping estimate can be made using half the track width and the CG height.

The half-track width is:

$$\frac{W_{track}}{2} = \frac{1.8}{2} \quad (42)$$

$$\frac{W_{track}}{2} = 0.9 \text{ m} \quad (43)$$

Assuming:

$$z_{CG} = 0.7 \text{ m} \quad (44)$$

the approximate lateral tip angle is:

$$\theta_{tip} = \tan^{-1} \left(\frac{W_{track}/2}{z_{CG}} \right) \quad (45)$$

$$\theta_{tip} = \tan^{-1} \left(\frac{0.9}{0.7} \right) \quad (46)$$

$$\theta_{tip} \approx 52.1^\circ \quad (47)$$

This is only a simplified static estimate and does not include dynamic effects, uneven terrain, wheel sinkage, or payload shifting. However, it suggests that the selected track width provides a reasonable starting point for lateral stability.

10 CAD Block Model Screenshot Section

This section should include screenshots of the first basic CAD block model. The purpose is not to show final detail, but to show that the rover proportions, subsystem placement, wheelbase, track width, and rocker-bogie layout are physically reasonable.

10.1 Top View

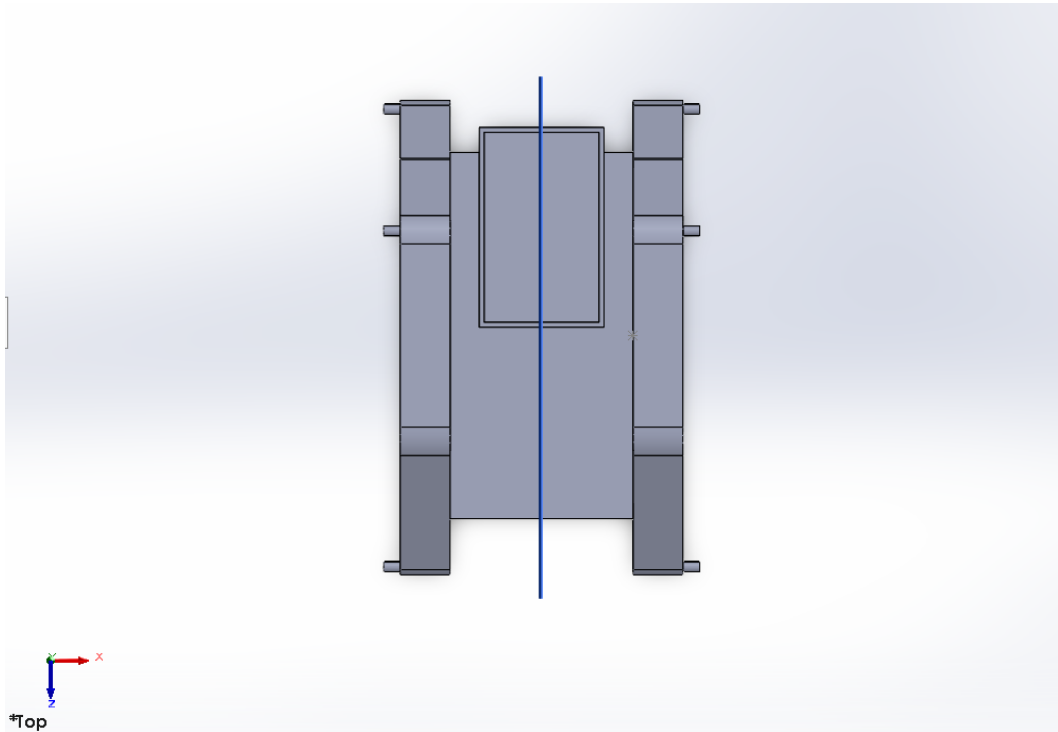


Figure 1: Top view of the preliminary rover block model showing overall length, track width, dump bed placement, and six-wheel layout.

10.2 Side View

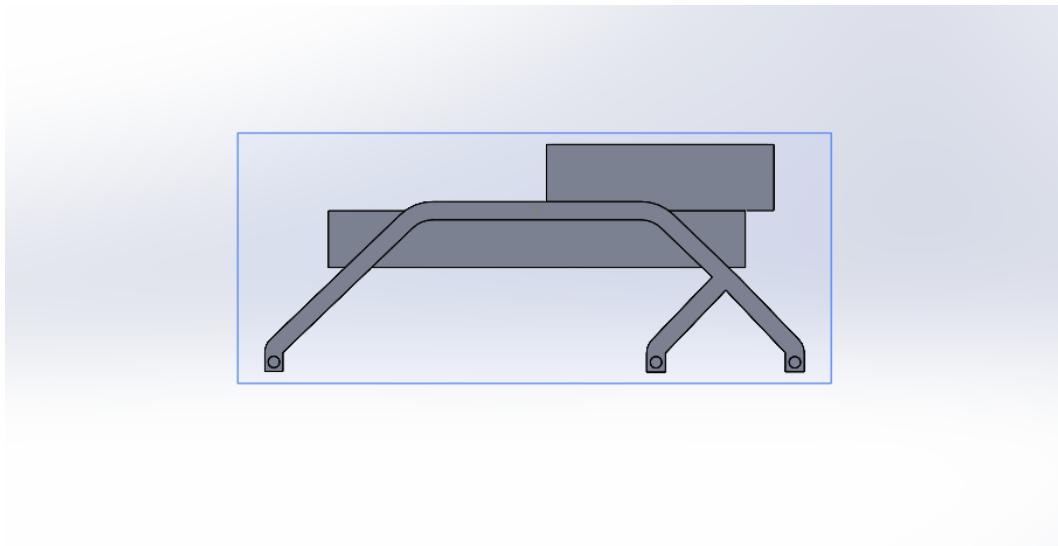


Figure 2: Side view of the preliminary rover block model showing rocker-bogie geometry, wheel spacing, ground clearance, chassis height, and dump bed position.

10.3 Isometric View

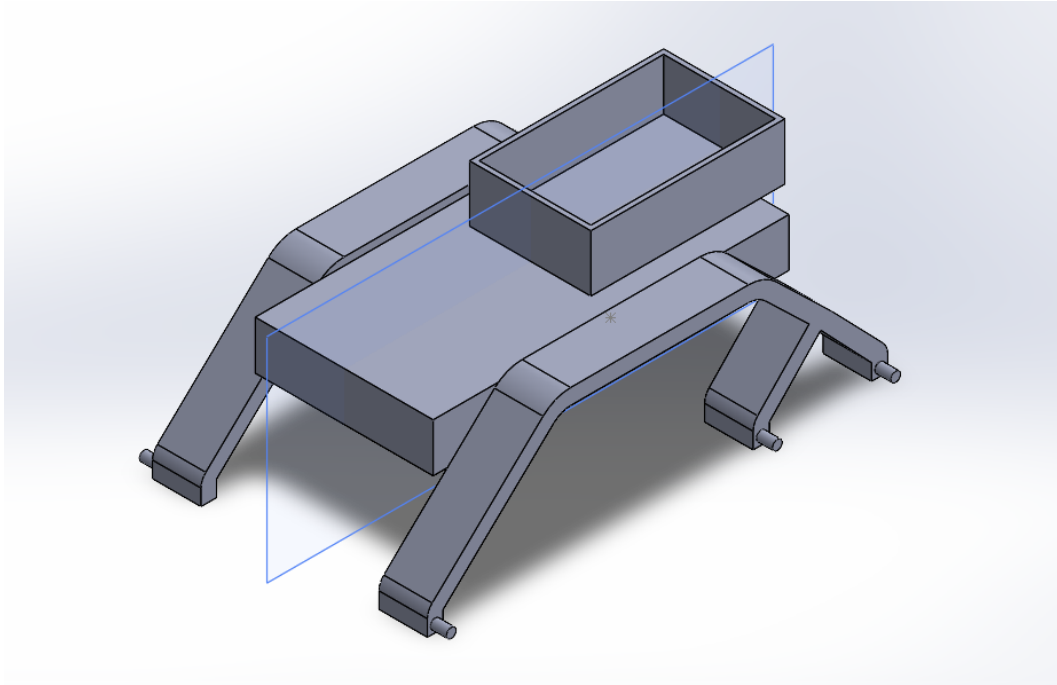


Figure 3: Isometric view of the rover block model showing the overall configuration of the chassis, dump bed, suspension, and wheel axles.

10.4 Rocker-Bogie Detail View

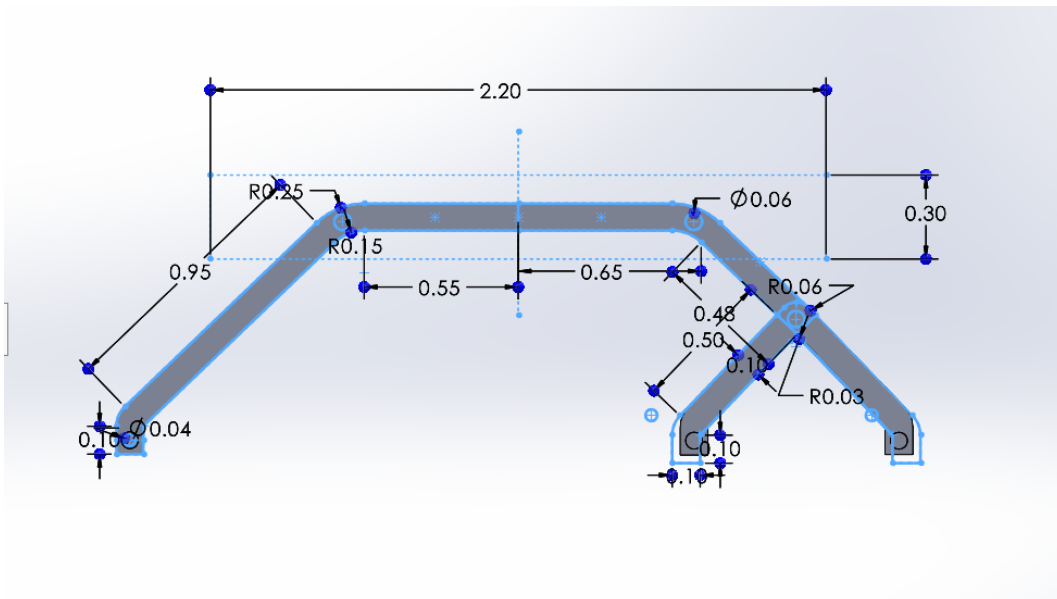


Figure 4: Close-up view of the preliminary rocker-bogie geometry, including the front wheel axle, middle wheel axle, rear wheel axle, rocker arm, bogie link, and pivot locations.

11 Week 4 Summary

This week finalized the rover's baseline payload and began converting the design from a concept sketch into a physically sized engineering model. The rover is now based around a 340 kg regolith payload and an estimated dump bed volume of approximately 0.30 m^3 . A preliminary dry mass of 650 kg was estimated, giving a loaded mass of approximately 990 kg. For early calculations, the loaded mass is rounded to 1000 kg. The first CAD block model will use an overall rover length of approximately 2.8 m, a track width of approximately 1.8 m, and six wheels with a diameter of approximately 0.50 m. The rocker-bogie system will be represented using simplified links and pivots before detailed joints, bearings, and motor housings are added.

Initial center-of-gravity thinking suggests that the battery and payload should be kept low and centered in the rover body. This should help improve stability during driving, turning, slope climbing, and dumping operations. Future work should refine the CAD model, calculate drivetrain torque requirements, evaluate slope climbing capability, and perform a more detailed center-of-gravity estimate using actual component locations.